

Arrest Data Visualization

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library(ggplot2)

## Warning: package 'ggplot2' was built under R version 4.5.2
setwd("~/Desktop/CSE160_Real_Estate/")

arrests_raw <- read.csv("NYPD_Arrest_Data_(Year_to_Date)_20251203.csv",
                       stringsAsFactors = FALSE)

# Use the original raw data for X/Y map
map_data <- arrests_raw

# Keep only valid boroughs
valid_boros <- c("M", "B", "K", "Q", "S")
map_data <- map_data[map_data$ARREST_BORO %in% valid_boros, ]

# Drop rows with missing or zero X/Y coordinates
map_data <- map_data[!is.na(map_data$X_COORD_CD) &
                     !is.na(map_data$Y_COORD_CD) &
                     map_data$X_COORD_CD != 0 & map_data$Y_COORD_CD != 0,]

map_data <- map_data[
  map_data$X_COORD_CD >= 900000 &
  map_data$X_COORD_CD <= 1060000 &
  map_data$Y_COORD_CD >= 120000 &
  map_data$Y_COORD_CD <= 275000,
]

# sample for speed
set.seed(160) # reproducible
n_keep <- min(200000, nrow(map_data)) # cap at 50k points
keep_idx <- sample(1:nrow(map_data), n_keep)
map_data <- map_data[keep_idx, ]

# Map borough abbreviations to full names (for legend)
borough_full <- map_data$ARREST_BORO
borough_full[borough_full == "M"] <- "Manhattan"
borough_full[borough_full == "B"] <- "Bronx"
borough_full[borough_full == "K"] <- "Brooklyn"
borough_full[borough_full == "Q"] <- "Queens"
borough_full[borough_full == "S"] <- "Staten Island"
map_data$BOROUGH <- factor(borough_full)
```

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# Plot: X_COORD_CD vs Y_COORD_CD, colored by borough
ggplot(map_data, aes(x = X_COORD_CD, y = Y_COORD_CD, color = BOROUGH)) +
  geom_point(alpha = 0.1, size = 0.5) +
  coord_equal() +
  theme_void() +
  theme(legend.position = "bottom") +
  scale_color_manual(
    values = c(
      "Bronx"          = "#E63946", # red
      "Brooklyn"       = "#2A9D8F", # green
      "Manhattan"      = "#264653", # dark blue
      "Queens"         = "#F4A261", # orange
      "Staten Island" = "#9D4EDD"  # purple
    )
  ) +
  labs(
    title = "NYPD Arrest Locations by Borough (X/Y Coordinate System)",
    color = "Borough"
  )

```

NYPD Arrest Locations by Borough (X/Y Coordinate System)



Borough • Bronx • Brooklyn • Manhattan • Queens • Staten Island

```

map_data <- arrests_raw

# Keep valid lat/long
map_data <- map_data[
  !is.na(map_data$Latitude) &
  !is.na(map_data$Longitude) &
  !(map_data$Latitude == 0 & map_data$Longitude == 0), ]

```

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# Sample to speed up plotting
set.seed(160)
map_data <- map_data[sample(1:nrow(map_data), min(200000, nrow(map_data))), ]

ggplot(map_data, aes(x = Longitude, y = Latitude)) +
  geom_point(alpha = 0.1, size = 0.5, color = "steelblue") +
  coord_quickmap() +
  theme_void() +
  labs(title = "NYC Arrests by Latitude/Longitude")

```

NYC Arrests by Latitude/Longitude



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heat_data <- arrests_raw

# Keep only valid boroughs
valid_boros <- c("M", "B", "K", "Q", "S")
heat_data <- heat_data[heat_data$ARREST_BORO %in% valid_boros, ]

# Drop rows with missing or zero X/Y coordinates
heat_data <- heat_data[
  !is.na(heat_data$X_COORD_CD) &
  !is.na(heat_data$Y_COORD_CD) &
  heat_data$X_COORD_CD != 0 &
  heat_data$Y_COORD_CD != 0,
]

# Keep only points inside NYC projected bounds
heat_data <- heat_data[
  heat_data$X_COORD_CD >= 900000 &

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heat_data$X_COORD_CD <= 1060000 &
heat_data$Y_COORD_CD >= 120000 &
heat_data$Y_COORD_CD <= 275000,
]

# Sample for speed
set.seed(160)
n_keep <- min(200000, nrow(heat_data))
heat_data <- heat_data[sample(1:nrow(heat_data), n_keep), ]

# Ranges for coord_equal (remove any remaining NA just in case)
x_range <- range(heat_data$X_COORD_CD, na.rm = TRUE)
y_range <- range(heat_data$Y_COORD_CD, na.rm = TRUE)

# Heatmap-style density of arrests in X/Y space
ggplot(heat_data, aes(x = X_COORD_CD, y = Y_COORD_CD)) +
  geom_bin2d(bins = 80) +
  coord_equal(xlim = x_range, ylim = y_range, expand = FALSE) +
  theme_void() +
  labs(
    title = "Density of NYPD Arrests (X/Y Coordinate Heatmap)",
    x = NULL,
    y = NULL,
    fill = "Count"
  )

```

Density of NYPD Arrests (X/Y Coordinate Heatmap)

